

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=cottage_cheese

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	nan	nan	nan	nan	nan	nan	ambiguous	0.0	cottage_cheese	silpo	promo_controlled	nan	nan
prozorro	1.1483227108295172e-10	0.01	4.589982408975803e-17	0.1	4.813752592554151e-10	0.1	I(1)	0.0	cottage_cheese	silpo	promo_controlled	nan	nan
retail	5.9157706543378735e-08	0.1	7.517018918246667e-13	0.0865089553841217	0.001856917684812617	0.0416666666666666796	I(0)	0.0	cottage_cheese	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	silpo	promo_controlled	nan	0.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	silpo	promo_controlled	0.0044991066582690566	35.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cottage_cheese	silpo	promo_controlled	nan	0.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
cottage_cheese	silpo	promo_controlled	daily	none	none	n/a	n/a	203	nan	nan	nan	nan	nan	nan	nan	unavailable	1	No admissible model estim...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
none	none	n/a	n/a	nan	nan	nan	1

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2025-04-14	nan	138.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-16	nan	118.8	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-18	nan	108.9	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-21	nan	120.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-22	nan	99.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-23	nan	107.08	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-24	nan	160.2	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-29	nan	156.06	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-04-30	nan	226.25	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-01	nan	82.5	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-02	nan	118.08000000000001	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-05	nan	101.29499999999999	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-06	nan	115.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-07	nan	258.2	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-08	nan	145.4	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-09	nan	101.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-11	nan	89.4	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-12	nan	168.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-13	nan	131.34	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-14	nan	75.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-15	nan	152.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-16	nan	145.2	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-19	nan	98.75	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-20	nan	159.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-21	nan	134.04	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-22	nan	139.98000000000002	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-23	nan	107.55	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-26	nan	168.6	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-27	nan	156.06	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-29	nan	150.5	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-05-30	nan	165.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-02	nan	135.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-03	nan	122.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-05	nan	142.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-06	nan	90.52000000000001	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-09	nan	60.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-10	nan	112.98	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-11	nan	93.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-12	nan	108.0	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...
2025-06-13	nan	71.5	nan	cottage_cheese	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
prozorro_to_retail	1	-0.03042410853195715	
prozorro_to_retail	2	0.09315059121833928	
prozorro_to_retail	3	-0.013554687740592215	
prozorro_to_retail	4	0.2640036394817208	
prozorro_to_retail	5	-0.01665792884614875	
prozorro_to_retail	6	0.12427459978543254	
prozorro_to_retail	7	-0.1940209713635291	
prozorro_to_retail	8	0.030775604018925283	
prozorro_to_retail	9	-0.09372737565447635	
prozorro_to_retail	10	-0.07704229808546917	
prozorro_to_retail	11	-0.17919361937712847	
prozorro_to_retail	12	0.18757301375096241	
prozorro_to_retail	13	0.10991817659598781	
prozorro_to_retail	14	-0.34943263469401786	
prozorro_to_retail	15	0.08988836742206224	
prozorro_to_retail	16	0.24534854931006433	
prozorro_to_retail	17	0.08079616663444002	
prozorro_to_retail	18	0.1868750222264715	
prozorro_to_retail	19	0.13127219217068242	
prozorro_to_retail	20	0.004357040410090344	
prozorro_to_retail	21	0.08986030260644852	
prozorro_to_retail	22	0.1322365935271666	
prozorro_to_retail	23	-0.3991469531086648	
prozorro_to_retail	24	0.16752752371267343	
prozorro_to_retail	25	0.05655494006971566	
prozorro_to_retail	26	-0.03561378213964211	
prozorro_to_retail	27	-0.000797420541569574	
prozorro_to_retail	28	0.20625749020049242	
prozorro_to_retail	29	0.06987978494683865	
prozorro_to_retail	30	0.055791968171944914	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
cottage_cheese	silpo	promo_controlled	daily	ambiguous	I(1)	I(0)	0	190	48		0	35	0 Low pair overlap in main ...

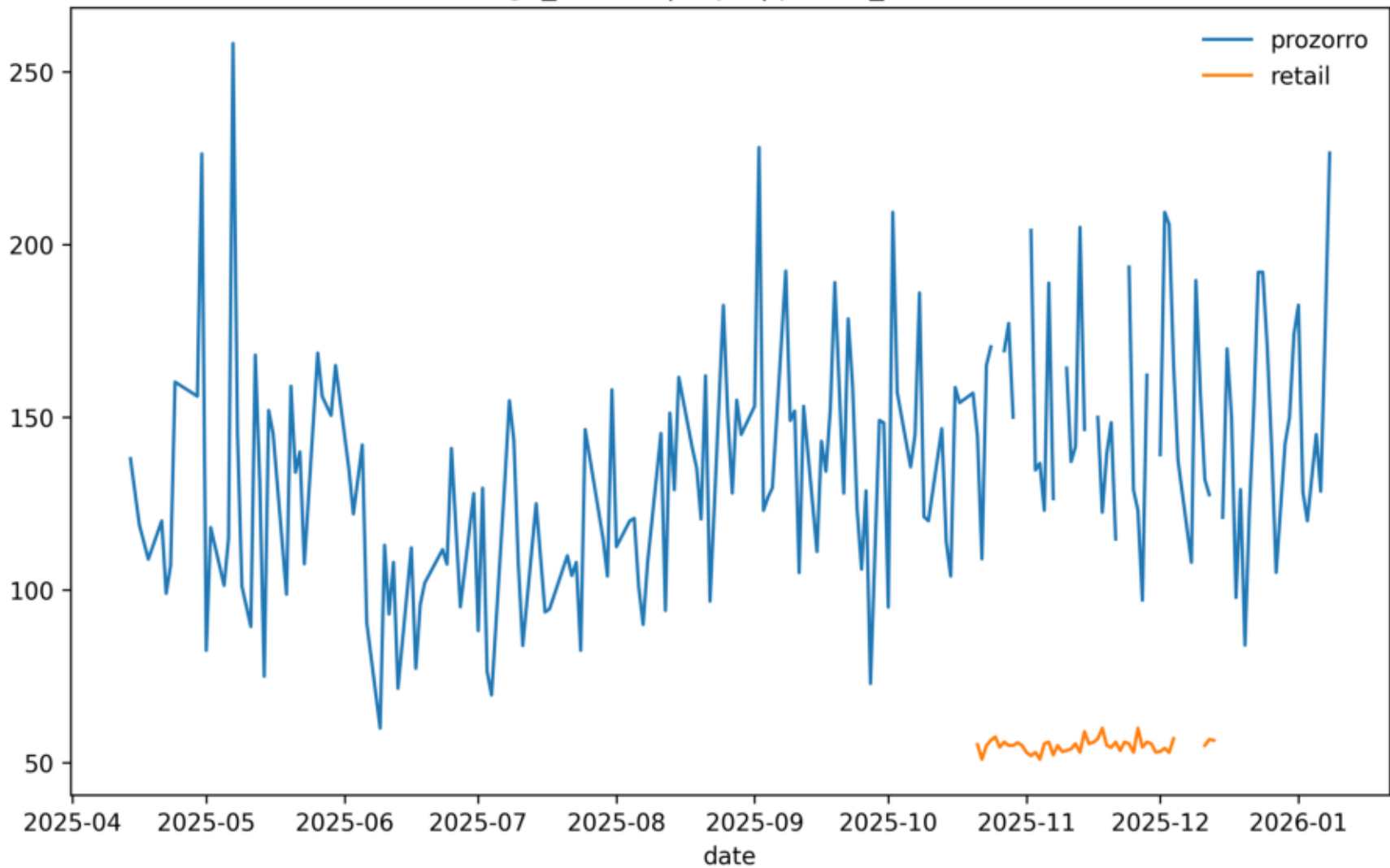
Table: NARDL_Multipliers

No rows

Table: VECM_IRF

No rows

time_series_promo_controlled.png
cottage_cheese | silpo | promo_controlled



Lag-Correlation Profile | cottage_cheese | silpo | promo_controlled

